Canonical Jacobi–Wright–Fisher Diffusions at Every Positive Mutation Pair: Boundary Atlas, Complete Spectrum, Heat Determinant, and Moment Closure

Route-A structural certificate C200

Abstract

We close one canonical dynamical atlas for the Jacobi diffusion at every positive mutation pair. The source lock includes the conservative no-flux realization and a generator twice the common one-half Wright–Fisher clock. Scale and speed classify every endpoint, including equality; the Beta law makes the generator reversible; shifted Jacobi polynomials give the complete spectrum, sharp gap, heat kernel, trace-class determinant and all moments. The positive-recurrent sample process revisits neighborhoods, whereas a positive-time semigroup fixes only constant observables. Exact independent certificates test these distinctions but do not replace the continuum proof. The family has no intrinsic prime arithmetic or target determinant and is strictly rejected by Route A.

Keywords: Jacobi diffusion; Wright–Fisher process; Beta law; boundary classification; Jacobi polynomials; heat kernel; recurrence.

中文摘要

本文对全部正突变参数下的典范Jacobi–Wright–Fisher扩散给出完整图谱。 源对象明确包含守恒零通量边界实现，且生成元时钟是常见二分之一约定的两倍。 尺度与速度密度确定全部端点类型；Beta分布给出可逆性， Jacobi多项式给出完整谱、热核、迹类行列式与全部矩。 样本路径正返并不矛盾于半群只有常值周期观测量。该系统没有内生素数算术，因此严格拒绝路线A。
关键词：Jacobi扩散；边界分类；可逆半群；谱隙；矩闭合。

1 Source lock and endpoint theorem

For $\alpha, \beta > 0$ let X be the conservative solution on $[0, 1]$ of

$$dX_t = [\alpha - (\alpha + \beta)X_t]dt + \sqrt{2X_t(1 - X_t)}\, dW_t, \tag{1}$$

with generator

$$Lf = x(1 - x)f'' + [\alpha - (\alpha + \beta)x]f'. \tag{2}$$

This is twice the clock in the neutral recurrent-mutation convention of Song and Steinrücken [2]; losing that factor halves every eigenvalue. When an endpoint is regular, the interior expression alone admits other extensions. We therefore freeze the canonical conservative/no-flux realization studied in the Wright–Fisher semigroup literature [1].

Theorem 1 (complete endpoint atlas). At zero the endpoint is regular reflecting for $0 < \alpha < 1$ and entrance for $\alpha \geq 1$. At one it is regular reflecting for $0 < \beta < 1$ and entrance for $\beta \geq 1$.

Proof. Up to positive constants the scale and speed densities are

$$s'(x) = x^{-\alpha}(1 - x)^{-\beta}, \qquad m'(x) = x^{\alpha-1}(1 - x)^{\beta-1}.$$

The speed integral is finite at zero for every $\alpha > 0$, while the scale integral is finite precisely when $\alpha < 1$. Feller's tests give regular versus entrance. The same calculation at one replaces α by β. The SDE/no-flux lock selects instantaneous reflection in the regular cases. In particular, equality belongs to the entrance side. □

2 Beta reversibility and the complete spectrum

Normalize the speed law to

$$\pi(dx) = B(\alpha, \beta)^{-1}x^{\alpha-1}(1 - x)^{\beta-1}\, dx.$$

Direct differentiation gives the divergence identity

$$Lf = \pi^{-1}\frac{d}{dx}\{x(1 - x)\pi f'\}, \qquad -\langle f, Lf\rangle_\pi = \int_0^1 x(1 - x)|f'|^2\, d\pi. \tag{3}$$

Thus the canonical closure is reversible and self-adjoint. Define

$$J_n(x) = P_n^{(\beta-1,\alpha-1)}(2x - 1), \qquad \lambda_n = n(n + \alpha + \beta - 1).$$

The Jacobi differential equation gives $LJ_n = -\lambda_n J_n$; orthogonal polynomial completeness gives an $L^2(\pi)$ basis [2, 3]. Therefore zero is simple, the sharp gap is $\lambda_1 = \alpha + \beta$, and

$$\|P_t f - \pi(f)\|_2 \leq e^{-(\alpha+\beta)t}\|f - \pi(f)\|_2.$$

3 Heat kernel, determinant and every moment

For the orthonormal Jacobi basis ϕ_n, the transition kernel relative to π is

$$k_t(x,y) = \sum_{n\geq 0} e^{-\lambda_n t}\phi_n(x)\phi_n(y). \tag{4}$$

Since $\lambda_n \asymp n^2$, P_t is trace class for every $t > 0$ and

$$\operatorname{tr} P_t = \sum_{n\geq 0} e^{-\lambda_n t}, \qquad \det(I - zP_t) = \prod_{n\geq 0}(1 - ze^{-\lambda_n t}). \tag{5}$$

This is the full-semigroup source determinant, including the stationary factor $1 - z$; it is not a primitive-orbit zeta or target divisor.

The monomial flag is invariant because

$$Lx^k = k(k+\alpha-1)x^{k-1} - k(k+\alpha+\beta-1)x^k. \tag{6}$$

Hence, with $m_0 = 1$,

$$\dot{m}_k = k(k+\alpha-1)m_{k-1} - k(k+\alpha+\beta-1)m_k, \qquad m_k^* = \frac{(\alpha)_k}{(\alpha+\beta)_k}.$$

This is an exact triangular solution algorithm at every degree, not a moment closure approximation.

4 Two meanings of recurrence

The finite invariant speed measure and irreducibility make the sample process positive recurrent: interior neighborhoods are revisited. Thus the blanket statement "there is no recurrence" is false. A different, exact semigroup statement follows from the spectrum. If $P_T f = f$ for $T > 0$ and $f = \sum c_n\phi_n$, then $(1 - e^{-\lambda_n T})c_n = 0$. Since $\lambda_n > 0$ for $n \geq 1$, only c_0 survives. Therefore the periodic L^2 observables are exactly the constants. Sample recurrence and absence of nonstationary semigroup cycles coexist.

5 Independent certificate and Route-A stop

Nine rational parameter pairs cross the two threshold equalities. Degrees zero through eight give 81 exact monic eigenpolynomials; stationary moments and the identity $HC = C^T H$ for the monomial Gram and generator matrices bring the ledger to 1,206 exact scalar identities. A standard-library checker imports no producer code and enforces every nested key set. A separate SymPy path, byte replay, fifteen repaired-hash attacks and one stale-hash attack all pass. This finite ledger tests formulas and clock conventions; the preceding proof establishes the continuum theorem.

There is no rational-prime carrier, deterministic primitive-orbit ledger, prime-power repetition or $\log p$ clock. The heat determinant in (5) has no target match. Although $-L$ is self-adjoint, replacing e^{tL} by $e^{is(-L)}$ is an imaginary-time change, not a same-clock quantization. Strictly,

$$(A0, A1, A2, A3, A4) = (\mathrm{FAIL}, \mathrm{FAIL}, \mathrm{FAIL}, \mathrm{FAIL}, \mathrm{FORMAL\ HINT}),$$

overall `ROUTE_A_REJECTED`, Route B false, under `NO_BAD_EULER_OR_ROOT_NUMBER`.

Claim firewall. We claim no classical priority, biological inference, target zero or prime data, arithmetic local datum, Euler factor, root number, automorphy, target functional equation or counting law, Weil compression, Hilbert–Pólya operator, external review or acceptance score.

Revision focus. Round 2 adds independent validation, hostile controls, source ownership and the strict Route-A firewall.

Declarations

Data and code availability. Package-local synthetic exact evidence and deterministic code accompany the paper; no biological observations are used.

Ethics. No human, animal, personal, clinical or sensitive data are used; approval is not applicable.

Author contributions (CRediT). This anonymous certificate records Conceptualization, Formal analysis, Software, Validation, Writing—original draft, and Writing—review and editing. AI systems are not authors.

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AI-use disclosure. An AI coding assistant supported drafting and exact-code development; it was not an external reviewer or independent error process.

References

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